Inteligență Artificială

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Fairness in AI

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- Useful resources
  - Kate Crawford, “The Trouble with Bias”,NeurIPS 2017 Keynote https://www.youtube.com/watch?v=fMym_BKWQzk
  - Timnit Gebru and Emily Denton, CVPR 2020 Tutorial on FATE/CV https://sites.google.com/view/fatecv-tutorial/schedule?authuser=0
  - Movie “The Social Dilemma” https://www.imdb.com/title/tt11464826/?ref_=vp_close
  - S. Borocas, M. Hardt, A. Narayanan, Fairness and ML, 2023 https://fairmlbook.org/
  - C. Molnar, *A Guide for Making Black Box Models Explainable, 2024* https://christophm.github.io/interpretable-ml-book/

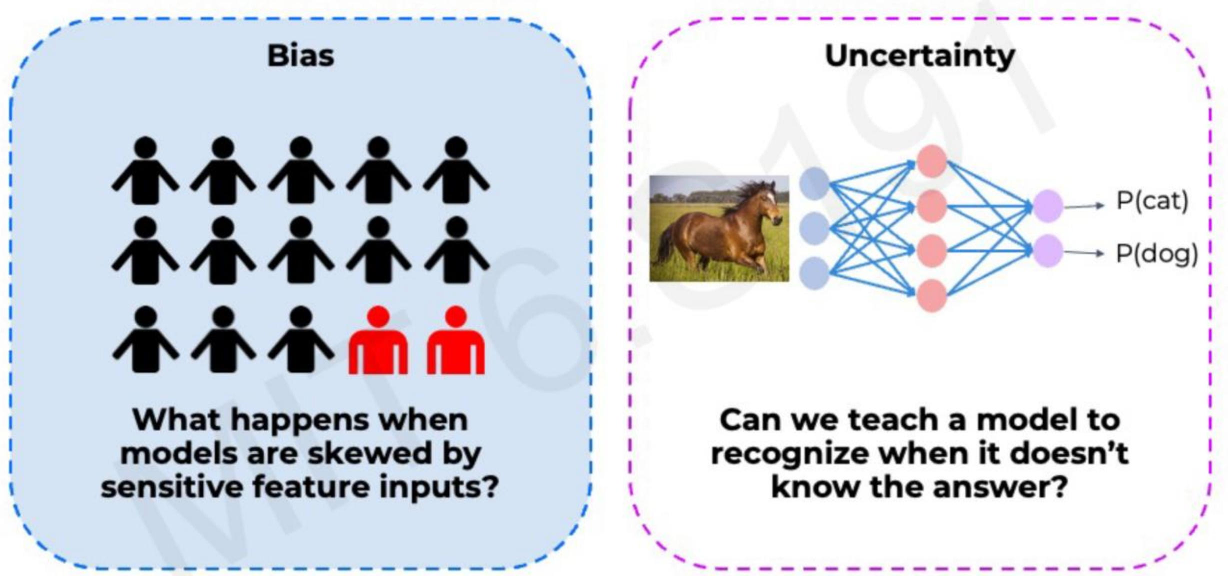
About trust

- Challenges in robust, safe and trustworthy ML

  - Interpretability – why?
  - Fairness – for all subgroups?
  - Robustness – for a large variety of data?
  - Uncertainty -

About trust

- Challenges in robust, safe and trustworthy ML
  - Bias
  - Uncertainty



Bias

- What?
  - When ML models do better on some demographics than others
- Bias in AI/ML lifecycle
  - Data
    - E.g. Sampling bias – over-sampling from some regions of the input data distribution and under-sampling from other regions
      - e.g. health data (a few diseased patients, a lot healthy patients)
    - E.g. Selection bias – proportion of data in dataset does not reflect the real-world
      - E.g. voice recognition for English
  - Model
    - Lack of uncertainty benchmarks and metrics
  - Deployment
    - Distribution shifts and feedback loops
  - Evaluation
    - Subgroups that are not or un-represented in the evaluation dataset
  - Interpretation
    - Human errors

Bias

- common human biases that can manifest in the data:
  - Reporting bias → strong opinions for books' review
  - Historical bias → house prices from '60-'80
  - Automation bias → ground-truth validation for tooth implants
  - Selection bias
    - Coverage bias → questionaires just for some groups (e.g. females & yogurt)
    - Participation-bias → different responding rate for different sub-groups
    - Sampling-bias → first 200 answers that are more enthusiastic
  - Implicit bias → gesture recognition and cultural particularities
  - experiment bias → train the model in a particular setup
- How to identify the bias?
  - Missing values for sub-groups of examples
  - Unexpected values
  - Data skew
- Mitigating the bias
  - Augmenting the training data
  - Adjusting the loss function

Bias

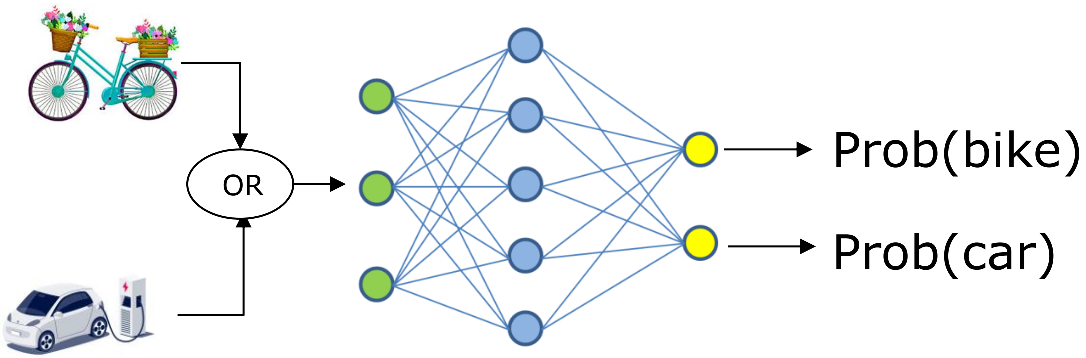
- How to evaluate the bias?
  - Demographic parity
  - Equality of opportunity

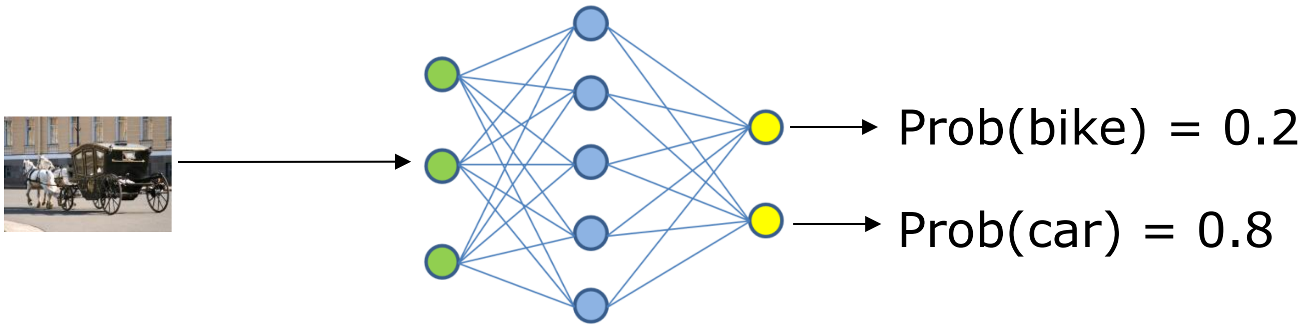
	Majority		Minority	
	accepted	rejected	accepted	rejected
	16	64	4	16
Acceptance rate	16 / 64 ~ 25%		4 / 16 ~ 25%	

	Majority		Minority	
	accepted	rejected	accepted	rejected
Qualified	16	19	4	11
Unqualified	0	45	0	5
Acceptance rate	16 / (16 + 19) ~ 46%		4 / (4 + 11) ~ 27%	

Model uncertainty

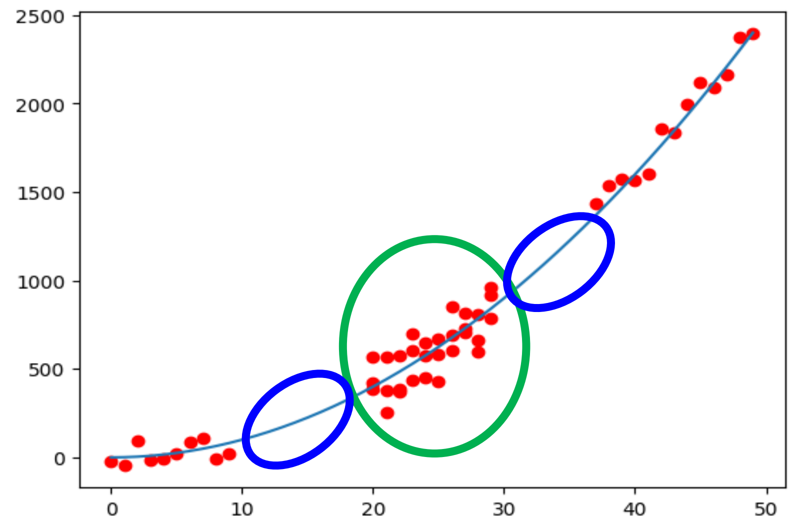
- Can we teach a model when it does or doesn't know the answer?
- Uncertainty estimation gives us a measure of confidence in the prediction



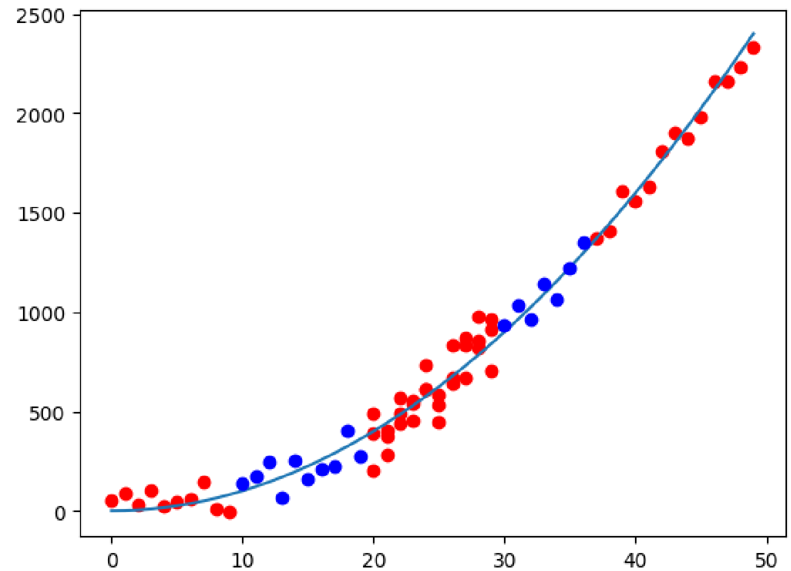


Types of uncertainity

- Data uncertainty
  - Similar inputs & very different outputs
- Model uncertainty
  - Inputs out of distribution



- What happens if we add new data?
  - Model uncertainty is reduced



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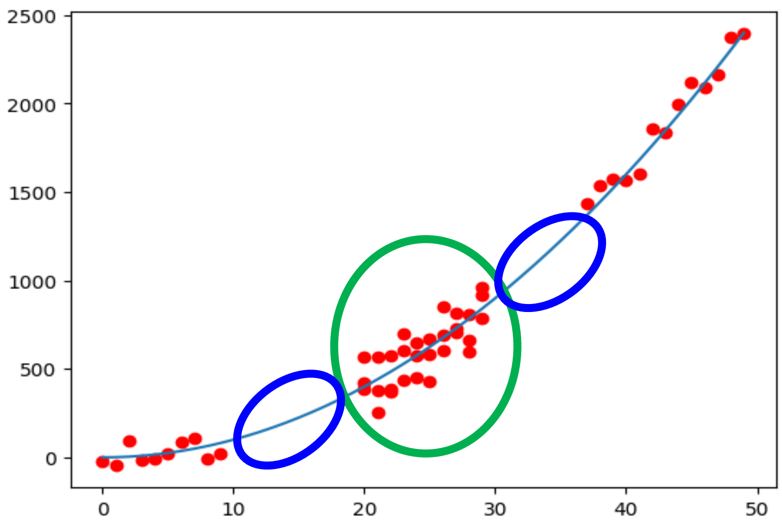
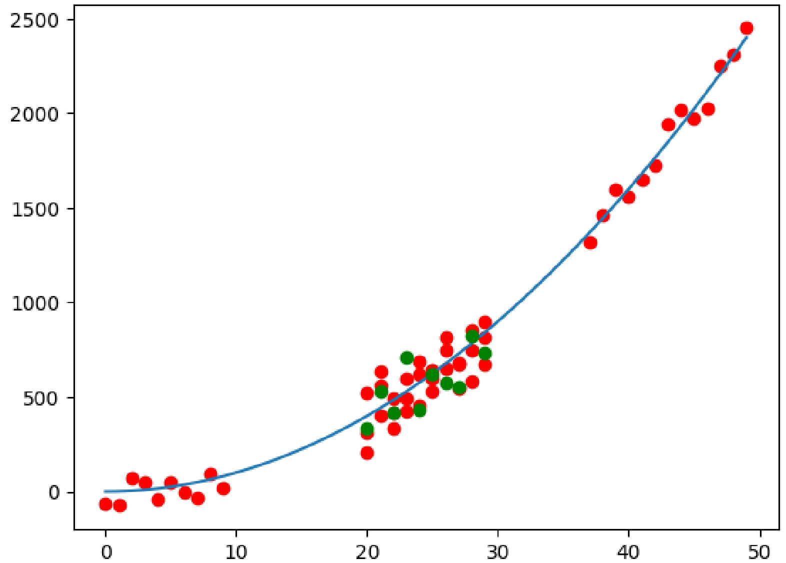
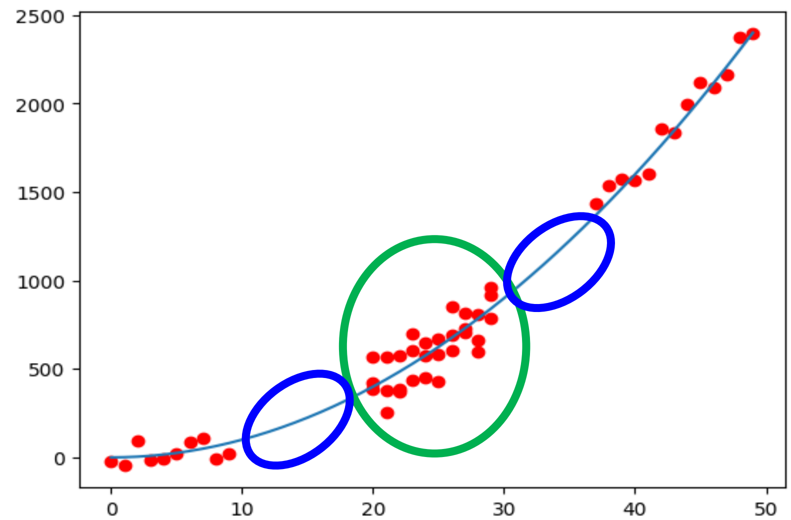


- What happens if we add new data?
  - data uncertainty is irreducible



Types of uncertainity

- Data uncertainty
  - Aleatoric uncertainty (intrinsic uncertainty)
  - Irreducible!!!
  - Can be learnt from data



- Model uncertainty
  - Epistemic uncertainty (uncertainty due to limitations in our knowledge about the world)
  - Can be reduced by adding data / knowledge
  - Cannot be learnt from data

Types of uncertainity

- Gave a loan to an individual, but they failed to repay; why?

- Data uncertainty
  - Aleatoric uncertainty (intrinsic uncertainty)
  - They were robbed

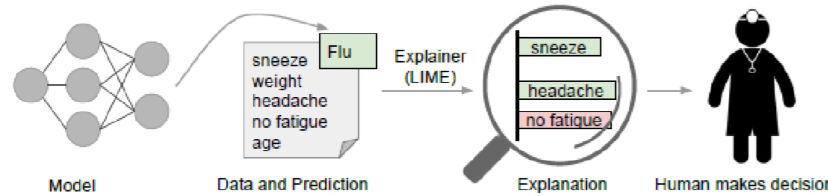
- Model uncertainty
  - Epistemic uncertainty (uncertainty due to limitations in our knowledge about the world)
  - They had bad credit score, but we didn’t bother checking

Explainability

- What?
  - Explanation = interpretable description of the model behavior
- Aim
  - Explain why the model made a particular prediction on a specific input
- How?
  - Select/rank a subset of input features that contributed the most to model prediction → Feature Attribution Methods

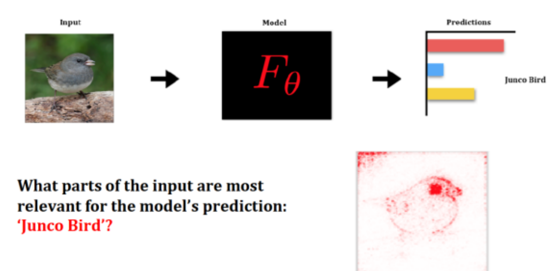
Explainability

- Approaches
  - LIME algorithm: “Why should I trust you?” Explaining the predictions of any classifier - Local Interpretable Model-agnostic Explanations https://github.com/marcotcr/lime



  - SHAP = SHapley Additive exPlanations - method based on cooperative game theory https://github.com/shap/shap

  - Saliency Maps (e.g. for images) https://christophm.github.io/interpretable-ml-book/pixel-attribution.html



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Limitations and new frontiers

”What I cannot create, I cannot understand.”
R. Feynman